A6061-RAM2 Thin Wall Component Measurements

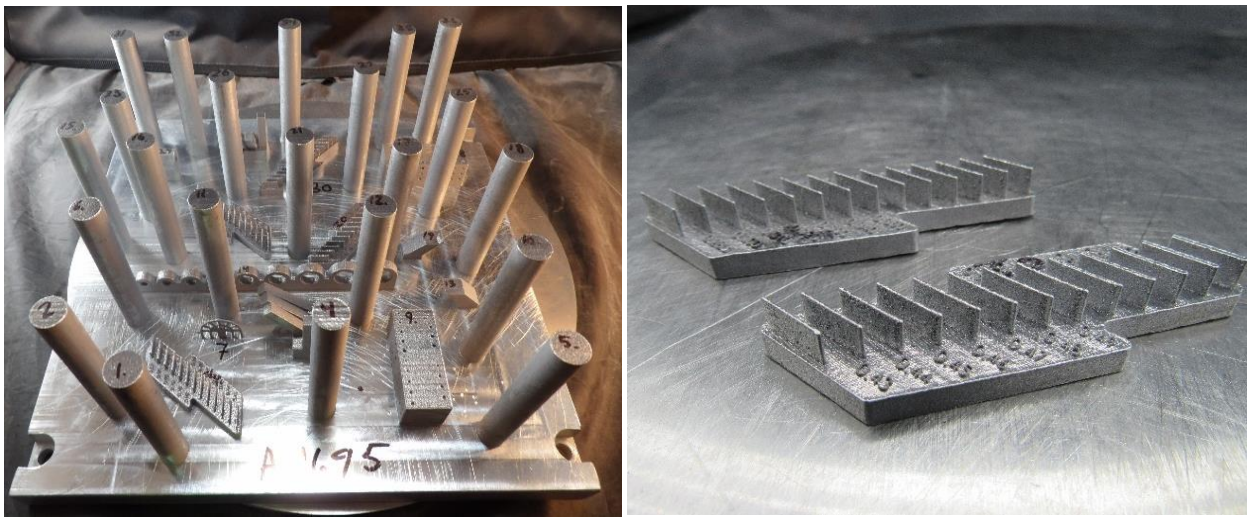


Figure 1: Left: build plate showing location and orientation of thin wall specimens. Recoating direction is right to left, argon flow from top to bottom. Right: Image of thin wall coupons, each printed in a different orientation.

Thin wall coupons of A6061-RAM2 using Elementum 3D's 30 µm settings were printed in two orientations, one sample where the flat surface of the thin walls faced the recoater and one where the flat walls were perpendicular to the recoater. Build layout and sample orientation, as well as the as built thin wall samples, are shown in **Figure 1**. After printing, all wall thicknesses were measured and compared to their nominal thickness. These results are shown in **Table 1**. All measurements were within Elementum 3D's geometric tolerance of 0.127 mm for printed components.

Table 1: Measurements from thin wall A6061-RAM2 components built in different orientations.

	Flat wall facing recoater		Flat wall perpendicular to reacoater	
Nominal Thickness (mm)	Average Thickness (mm)	Difference (Average-nominal) (mm)	Average Thickness (mm)	Difference (Average-nominal) (mm)
0.30	0.342	0.059	0.384	0.084
0.35	0.422	0.072	0.392	0.042
0.40	0.456	0.056	0.402	0.002
0.45	0.520	0.070	0.466	0.016
0.50	0.596	0.074	0.574	0.074
0.60	0.699	0.099	0.685	0.085
0.70	0.805	0.105	0.792	0.092
0.80	0.901	0.101	0.880	0.080
0.90	1.002	0.102	0.968	0.068
1.00	1.102	0.102	1.07	0.066